

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 2

CANDIDATE
NAME

CLASS

BIOLOGY

9744/01

Paper 1 Multiple Choice

20 September 2022

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name and CT on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

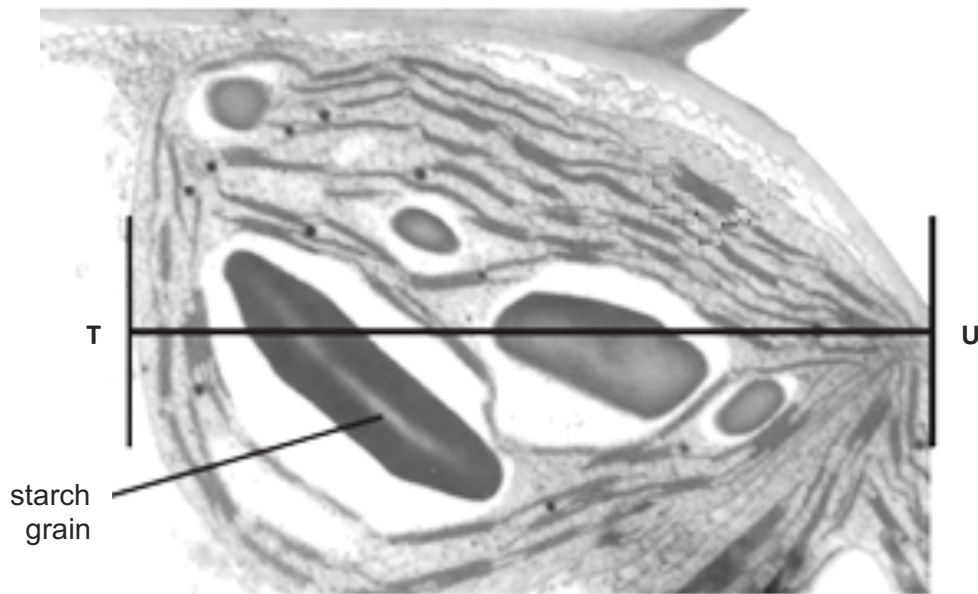
Calculators may be used.

This document consists of **18** printed pages and **0** blank pages.

[Turn over

2

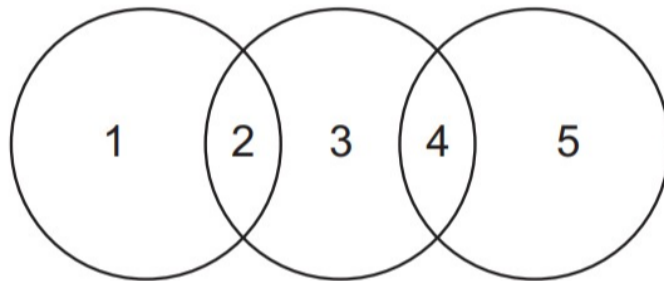
- 1 The electron micrograph shows part of a chloroplast in a plant cell. The actual length of the labelled starch grain in the chloroplast is $2.2\text{ }\mu\text{m}$ long.



What is the actual length of this chloroplast indicated by the line between T and U?

- A** 48 mm **B** $4.9\text{ }\mu\text{m}$ **C** 107 mm **D** $16.5\text{ }\mu\text{m}$

- 2 The diagram shows some similarities between chloroplasts, mitochondria and typical prokaryotes.



Which row is correct?

	1	2	3	4	5
A	chloroplasts	circular DNA	mitochondria	80S ribosomes	prokaryotes
B	chloroplasts	80S ribosomes	mitochondria	circular DNA	prokaryotes
C	prokaryotes	circular DNA	mitochondria	circular DNA	chloroplasts
D	prokaryotes	70S ribosomes	chloroplasts	80S ribosomes	mitochondria

- 3 A student carried out the Benedict's test on three different solutions, X, Y and Z. All three solutions have the same concentration, but contain different carbohydrates.

A sample of each solution was incubated with acid for five minutes, and then heated in a boiling water-bath with Benedict's solution. The time taken for the first appearance of a colour change was recorded.

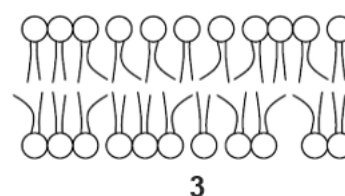
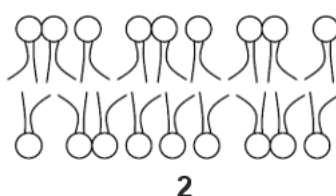
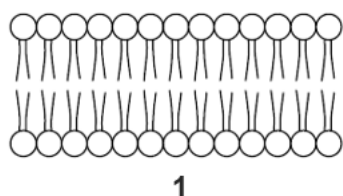
The results are shown in the table below.

solution	time for first appearance of a colour change with Benedict's solution / s
X	14
Y	300
Z	28

Which row correctly identifies the carbohydrate present in each solution?

	X	Y	Z
A	glucose	starch	maltose
B	glucose	sucrose	lactose
C	lactose	starch	glucose
D	lactose	sucrose	maltose

- 4 Cell membranes in three different organisms adapted for different climates are shown below.



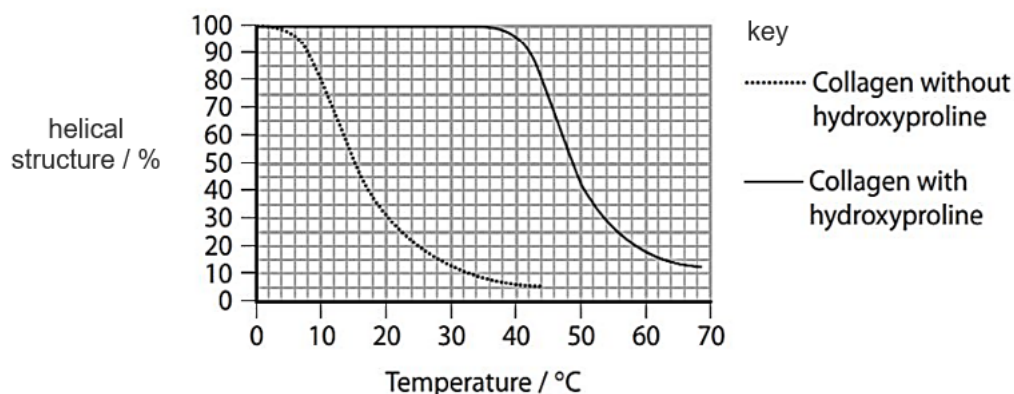
Several regions of the world are known to have the following climates:

- desert region is hot and dry
- tropical region is warm and wet
- temperate region is cool and moderate
- arctic region is cold and dry

Which of the following correctly matches the regions to each type of cell membrane?

	1	2	3
A	desert	arctic	temperate
B	arctic	desert	tropical
C	desert	temperate	arctic
D	tropical	arctic	desert

- 5 Hydroxyproline and proline are components of collagen. The graph shows the effect of temperature on the helical structure of collagen with and without hydroxyproline.



Which statements explain the importance of hydroxyproline in the structure of collagen?

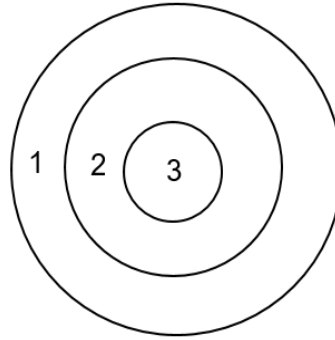
- 1 Presence of hydroxyproline increases thermal stability of collagen.
 - 2 At higher temperature, collagen with hydroxyproline results in less helices present than collagen without hydroxyproline.
 - 3 Hydroxyproline contribute to the extensive hydrogen bonds that hold the helices together.
- A** 1 and 2 **B** 1 and 3 **C** 2 and 3 **D** 1, 2 and 3
- 6 Under normal metabolism, alcohol is broken down in the liver by the enzyme alcohol dehydrogenase to acetaldehyde, which is then converted by the enzyme acetaldehyde dehydrogenase to a harmless acetic acid derivative.

Disulfiram is a drug used to support the treatment of alcoholism. It is a specific enzyme inhibitor of acetaldehyde dehydrogenase. In the presence of disulfiram, the Michaelis-Menten constant (K_m) for acetaldehyde dehydrogenase increases. In the presence of disulfiram, alcohol consumed can only be partly broken down, resulting in nausea and vomiting.

Why is it important that the enzyme inhibitor disulfiram is specific?

- 1 It cannot disrupt other metabolic pathways.
 - 2 It prevents alcohol from binding to the active site.
 - 3 It is unlikely to cause unwanted side effects.
 - 4 It binds to an allosteric site of acetaldehyde dehydrogenase.
- A** 1, 2, 3 and 4 **B** 1, 2 and 3 only **C** 1 and 3 only **D** 2 and 4 only

7 The diagram shows the relationship between the different potency of stem cells.



Which of the following options describes the potency of stem cells found in 1 - 3?

	1	2	3
A	differentiate into any cell type to form the whole organism	differentiate into limited range of cells and tissues appropriate to their location	differentiate into almost any cell type to form any organ or type of cell
B	differentiate into limited range of cells and tissues appropriate to their location	differentiate into almost any cell type to form any organ or type of cell	differentiate into any cell type to form the whole organism
C	differentiate into any cell type to form the whole organism	differentiate into almost any cell type to form any organ or type of cell	differentiate into limited range of cells and tissues appropriate to their location
D	differentiate into almost any cell type to form any organ or type of cell	differentiate into any cell type to form the whole organism	differentiate into limited range of cells and tissues appropriate to their location

8 The mechanism of action of four drugs that inhibit DNA replication is stated below.

- Aphidicholine inhibits DNA polymerase.
- Cytarabine is converted into a molecule that can substitute for a DNA nucleotide and also inhibits DNA repair mechanisms.
- Epirubicin inhibits an enzyme involved in the unwinding of DNA and separation of strands.
- Hydroxycarbamide inhibits an enzyme involved in the production of deoxyribonucleotides.

Which row correctly matches a drug to an explanation of the mechanism of action?

	explanation of mechanism of action			
	decreased pool of available nucleotides inhibits chain elongation	DNA strands not available as templates for transcription	DNA damaged during replication and cell death occurs	exposed DNA template strands unable to be copied
A	aphidicholine	epirubicin	cytarabine	hydroxycarbamide
B	epirubicin	cytarabine	hydroxycarbamide	aphidicholine
C	hydroxycarbamide	aphidicholine	epirubicin	cytarabine
D	hydroxycarbamide	epirubicin	cytarabine	aphidicholine

9 The following events occur during transcription.

- 1 Bonds break between complementary bases.
- 2 Bonds form between complementary bases.
- 3 Sugar-phosphate bonds form.
- 4 Free nucleotides pair with complementary nucleotides.

Before the mRNA leaves the nucleus, which events will have occurred twice?

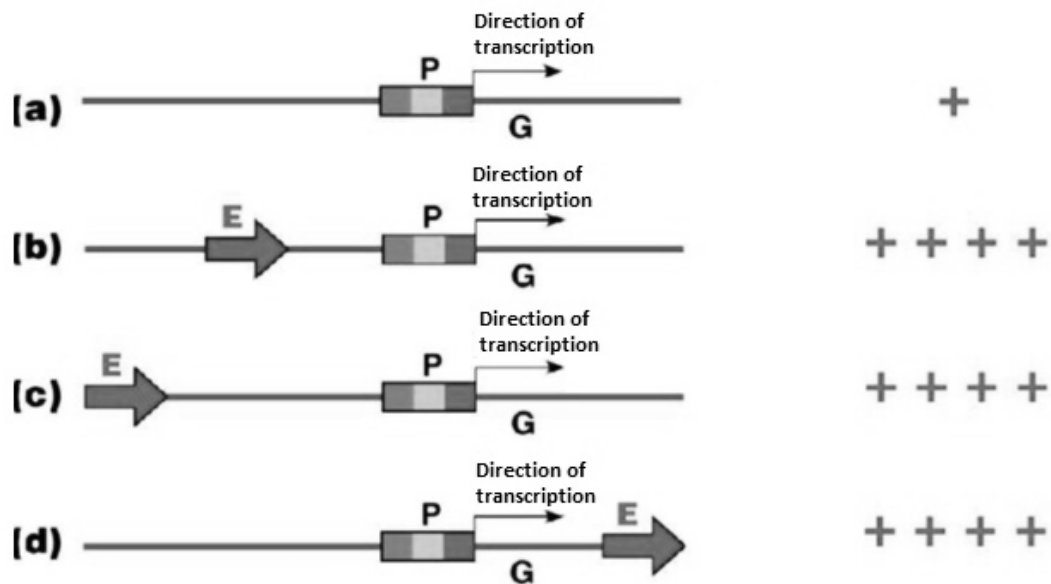
- A** 1 and 2 only
- B** 1, 3 and 4 only
- C** 2, 3 and 4 only
- D** 1, 2, 3 and 4

10 Which is **not** a description of a gene?

- A** a length of DNA which carries coded information as a sequence of nucleotides that can result in the formation of a polypeptide chain
- B** any section of a molecule that has two strands, each with a sequence of nucleotides that are complementary to each other and are held together by hydrogen bonding
- C** a sequence of nucleotides which can be copied by complementary base pairing and then be translated at a ribosome
- D** a sequence of nucleotides that can be transcribed using a polymerase enzyme and free activated nucleotides and which results in the formation of a messenger RNA molecule

11 Recombinant DNA techniques can be used to alter the locations of control elements within DNA sequences, so as to study the effects of these changes on the levels of transcription. The diagram shows the various structures of transcription units and the corresponding relative levels of transcription.

The promoter, enhancer and coding sequences are represented by letters P, E and G respectively. The number of symbol '+' indicates the relative levels of transcription.



With reference to the diagram, which of these statements is a valid conclusion?

- A** The relative distance between promoter and enhancer has no effect on the rate of transcription.
- B** The rate of transcription is increased when the enhancer is located upstream of the promoter.
- C** An enhancer is required for transcription.
- D** A promoter bound with RNA polymerase and general transcription factors result in basal rate of transcription.

- 12 When the products of the lac operon are not made regardless of the presence or absence of lactose.

Which of the following statement(s) is/are possible reason(s)?

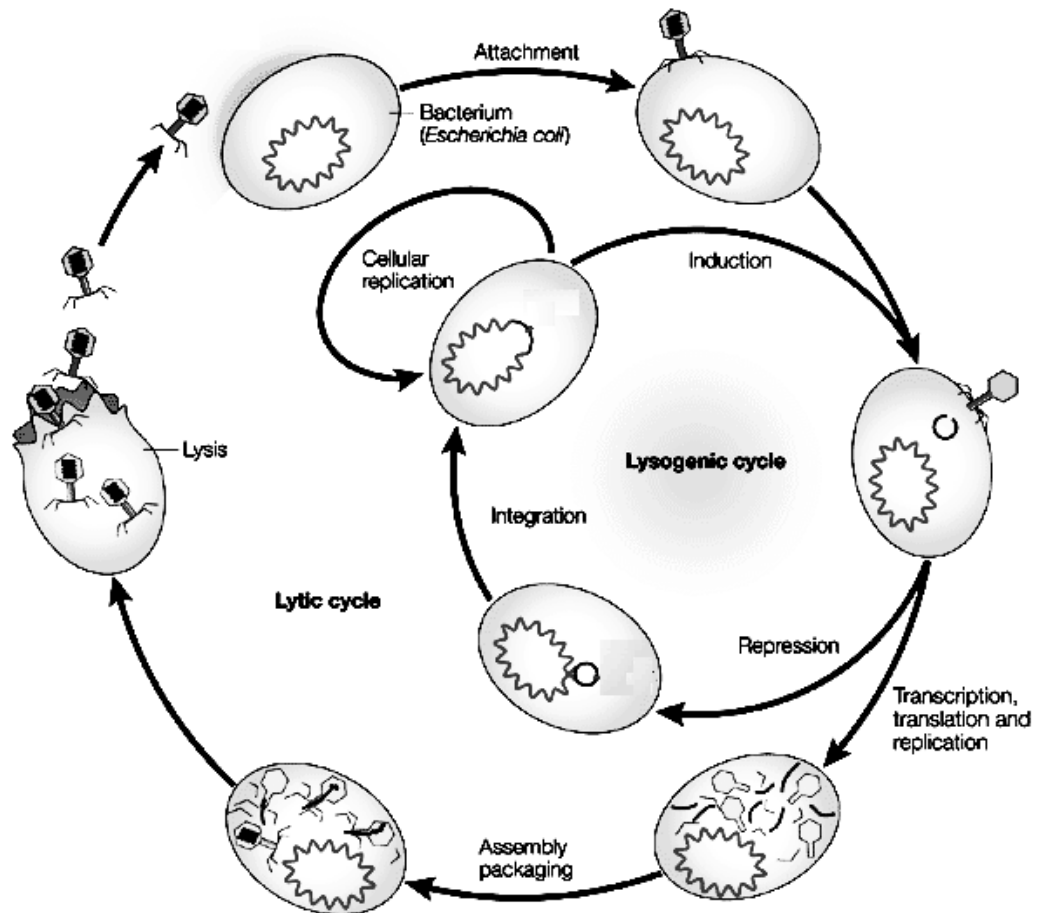
- 1 Mutation of the *lacI* regulatory gene resulting in a hyperactive protein.
- 2 Mutation of operator site.
- 3 Mutation of the RNA polymerase binding site
- 4 Insufficient amino acids in the bacteria.

- A 1 only
- B 1 and 4 only
- C 1, 3 and 4 only
- D 1, 2, 3 and 4 only

- 13 Which of the following statement comparing the human immunodeficiency virus (HIV) and lambda phage is **incorrect**?

- A The HIV enters by receptor-mediated endocytosis, but the lambda phage infects bacterial cells by injecting its DNA.
- B The capsid of the HIV enters the host cell, but the capsid of the lambda virus does not.
- C The genome of the HIV must be processed before it is integrated into the host chromosome, but the genome of the lambda virus can be directly integrated.
- D New HIV are released from the host cell via budding, but new lambda virus are released via cell lysis.

14 The diagram below shows the reproductive cycles of a phage infecting a host bacterium.



Some statements about the phages are listed as follows:

- 1 All types of phages are capable of undergoing lytic and lysogenic cycles.
- 2 A phage usually undergoes lysogenic cycle because a larger number of progeny phages can be produced rapidly as compared to the lytic cycle.
- 3 The release of lysozyme upon rupturing of lysosome leads to the osmotic lysis of host bacterium.
- 4 The phage gene codes for a repressor protein that prevents the expression of prophage.

Which of these statements about the reproductive cycles of a phage are **not** true?

- A 1 and 2 only
- B 3 and 4 only
- C 1, 2 and 3 only
- D 2, 3 and 4 only

15 Platypus is an egg-laying mammal with a diploid number of 52.

Which of the following information is correct?

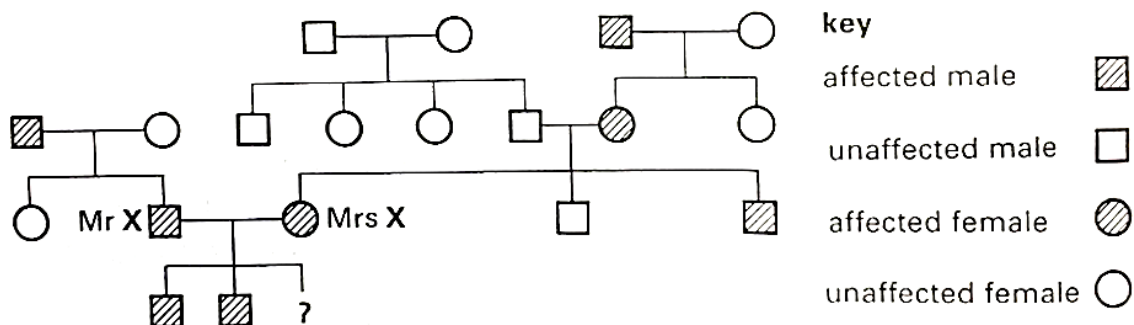
	number of centromeres during metaphase in a mitotic cell	number of chromatids in a cell at Anaphase I	number of DNA molecules in a cell at Anaphase II
A	26	52	26
B	26	104	52
C	52	104	52
D	52	104	26

16 Which of the following statements correctly describe the changes in cancer cells?

- 1 Limitless replicative potential often results in the accumulation of chromosomal mutations in many cancer cells.
- 2 Cancer cells could overproduce signal molecules so that they become self-sufficient in growth signals.
- 3 Angiogenesis is the result of expression of oncogenes in a cell line that produces blood vessels.
- 4 Loss-of-function mutations in tumour suppressor genes contribute to tissue invasion and metastasis.

- A** 1 and 4
B 2 and 3
C 1, 2 and 4
D 2, 3 and 4

17 The family tree below was assembled by a genetic counsellor for Mr and Mrs X who suffer from heart disease due to hypercholesterolaemia. Children who inherit the dominant mutant allele from both parents rarely survive beyond puberty.



What is the probability that Mr and Mrs X's third child will be unaffected?

- A** 0.75 **B** 0.50 **C** 0.25 **D** 0.00

- 18** The inheritance of fruit colour of eggplant is controlled at two different gene loci, E and F. The colour of eggplant fruit can be either purple or white.

An eggplant, heterozygous at both gene loci, is self-pollinated. Out of 30 offspring, 5 white and 5 purple offspring were randomly sampled. Their genotypes were determined and summarised in the table below.

eggplant fruit colour	genotypes
purple	EeFf, EEff, EeFF
white	eeFf, eeff, Eeff, EEff

What percentage of the eggplants will have the white phenotype?

- A** 7%
B 19%
C 44%
D 56%
- 19** The gene loci W, X, Y and Z forms a linkage group.

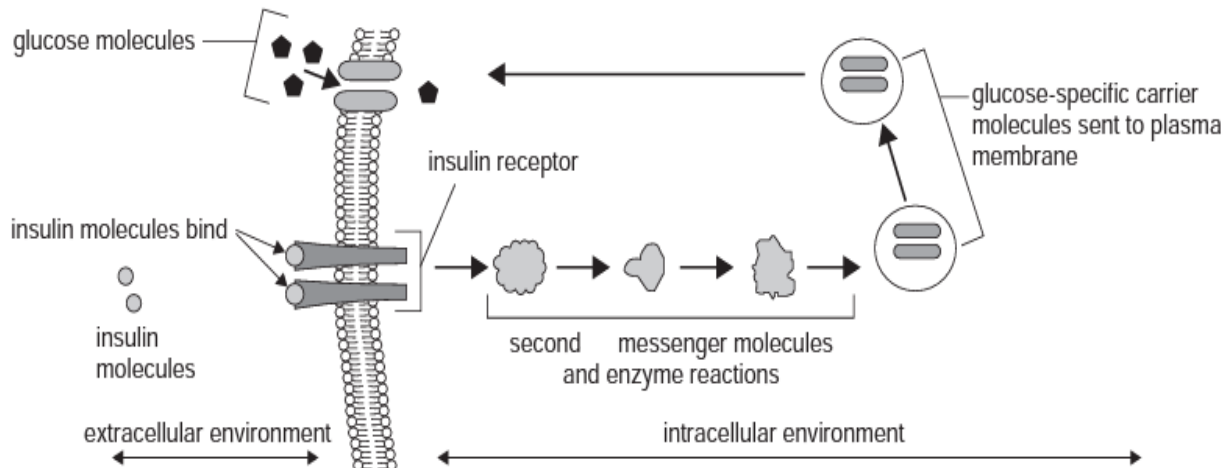
The table below shows the recombination frequencies when different pairs of gene loci in this linkage group were investigated.

pairs of gene loci	recombination frequency / %
W and X	46
W and Y	8
X and Y	52
X and Z	13
Y and Z	41

Which of the following shows the correct order of genes on the chromosome?

- A** W, X, Y, Z
B W, Y, Z, X
C Y, W, Z, X
D Y, Z, X, W

- 20** The diagram below shows a summary of the steps in an insulin signaling pathway that results in increased glucose uptake



A scientist studied the insulin signaling pathways of two female patients, Eleni and Shan.

Eleni's pathway is the same as that shown in the diagram above.

The scientist discovered that the gene that encodes the insulin receptor in Shan has a mutation. Insulin molecules cannot bind to Shan's insulin receptors.

From this information, it would be correct to conclude that

- A** insulin acts as a hydrophilic signaling molecule in Eleni and Shan.
- B** there would be more glucose-specific carrier molecules in Shan's plasma membranes than in Eleni's.
- C** the binding of insulin molecules to the receptor initiates transduction and the uptake of glucose into Eleni's cells.
- D** the presence of insulin in Shan would cause an increase in the concentration of the second messenger molecules.

- 21** How many of these statements about second messengers are true?

- 1 All second messengers are water-soluble protein molecules.
- 2 Second messengers pass the signal on by binding to and altering the behaviour of selected signalling or effector proteins.
- 3 Second messengers are generated in large amounts in response to receptor activation and diffuse away from their source, spreading the signal to other parts of the cell.
- 4 Second messengers may allow for signal amplification to occur.

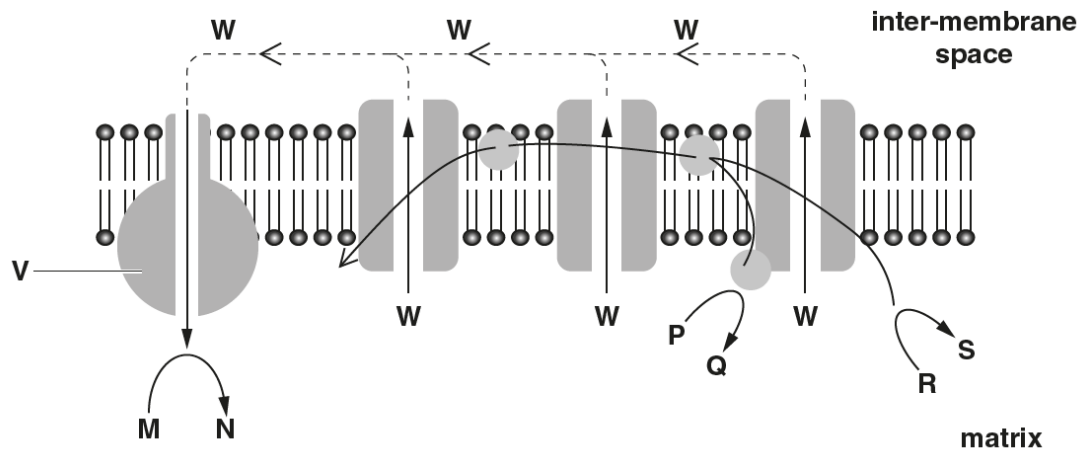
A 1

B 2

C 3

D 4

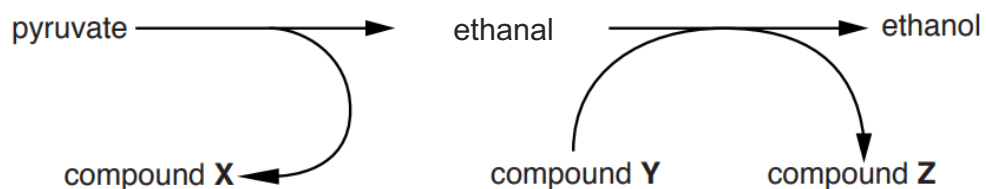
- 22** The diagram represents part of the inner membrane of a cell organelle. The letters represent substances involved in one stage of aerobic respiration.



Which row shows the possible combination of the substances?

	V	W	reduced coenzyme	ATP
A	ATP synthase	protons	P	N
B	ATP synthase	protons	S	M
C	H ⁺ channel	electron	R	N
D	H ⁺ channel	electron	S	M

- 23** The flowchart outlines the process of anaerobic respiration in yeast cells.



Which of the statements regarding the identities of the compounds are correct?

- Compound X is a waste product.
- Compound Y is necessary for more rounds of glycolysis.
- Compound Z acts as an oxidising agent.

- A** 1, 2 and 3
B 1 and 2 only
C 1 and 3 only
D 2 and 3 only

24 Different types of reaction occur in the sequence of chemical reactions known as the Calvin cycle.

Which reaction of the Calvin cycle is correctly described?

- A** Carboxylation occurs in the conversion of triose phosphate to RuBP.
- B** Decarboxylation occurs in the conversion of RuBP to GP.
- C** Phosphorylation occurs in the conversion of RuBP to GP.
- D** Reduction occurs in the conversion of GP to triose phosphate.

25 Two areas of molecular biology that have received considerable attention in evolutionary studies are the genetic code and cytochrome C. Cytochrome C is an essential component of all respiratory electron transport chains.

Which statements lend evidence to the ideas that

- all living organisms are related
 - there is a single, rather than a multiple, origin of life?
- 1 The almost universal nature of the genetic code is a result of evolutionary convergence from multiple lineages.
 - 2 The sequence of amino acids in cytochrome C is similar in organisms that are from similar environments or with similar metabolic demands.
 - 3 The majority of organisms have the same, or similar, amino acid sequences for cytochrome C.
 - 4 When transferred into a very dissimilar organism, a gene coding for cytochrome C will lead to the expression of a protein that will function in the other organism.

- A** 1 and 2 only
- B** 2 and 3 only
- C** 3 and 4 only
- D** 1, 3 and 4 only

- 26** Copies of a segment of DNA from species **X** were hybridised with copies of the homologous segment of DNA from species **P**.

The hybridised DNA segments were heated, and the temperature at which complete separation occurred (separation temperature) was recorded.

To determine evolutionary relationships, this process was repeated three times using DNA from species **Q**, **R**, and **S**. The results are shown in the table.

hybridised DNA	separation temperature / °C
P-X	30
Q-X	40
R-X	70
S-X	80

Which statement is supported by the data in the table?

- A** Species **P** and species **X** share a more distant common ancestor as they have the least percentage difference in their DNA.
- B** Species **P** and species **Q** together have the same percentage difference in DNA from species **X** as has species **R**.
- C** Species **R** is more closely related to species **S** than it is to species **X**.
- D** Species **S** and species **X** share a more recent common ancestor as they have the least percentage difference in their DNA

- 27** Barombi Mbo and Bermin are two volcanic crater lakes found 120km apart in Cameroon, Central Africa. They are extremely small, but harbor 11 and 9 endemic species of freshwater fishes known as cichlids respectively.

The cichlids could be differentiated by their different mating colouration. It was found that these individual cichlids with similar phenotypes mate with one another more frequently than would be expected under a random mating pattern.

Based upon genome analysis of the twenty species of cichlids, the cichlids found in Barombi Mbo and Bermin each formed a monophyletic group.

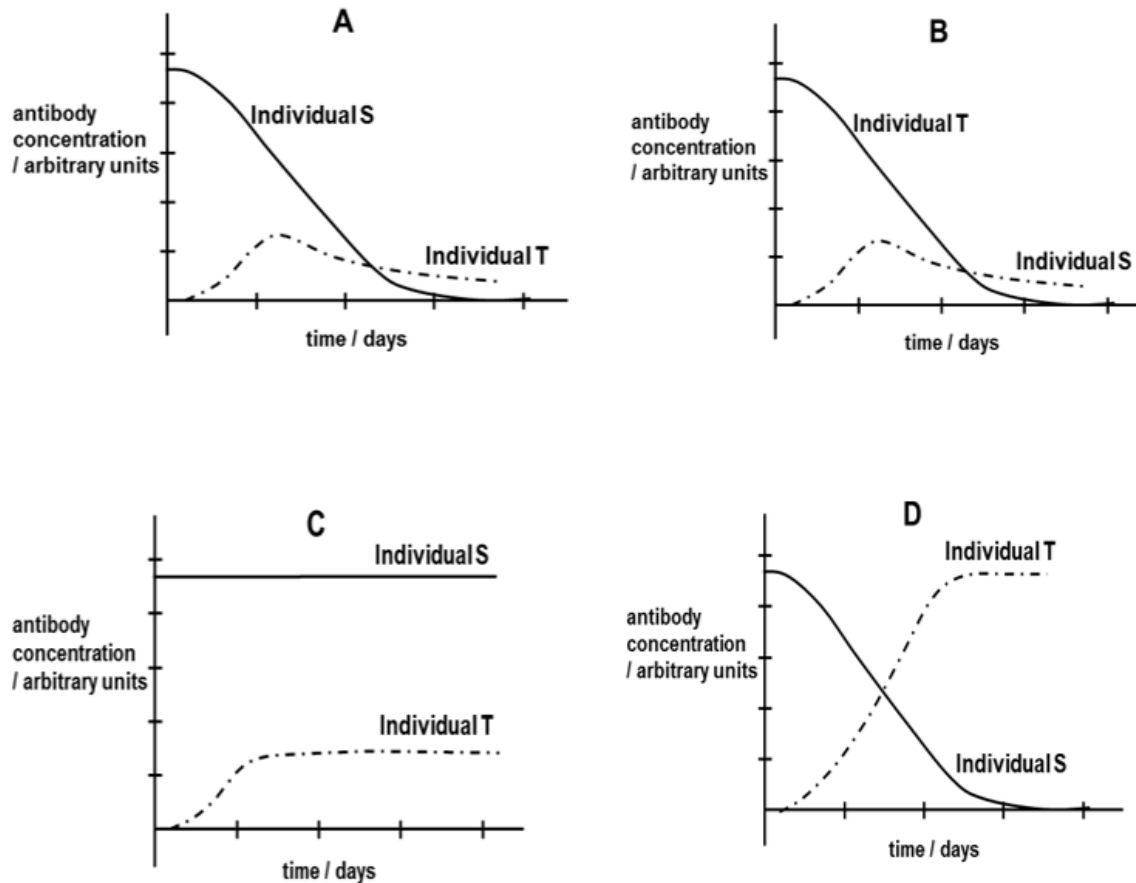
Which statements could apply to these cichlids?

- 1 Different species between Barombi Mbo and Bermin arose by allopatric speciation due to the presence of a geographical barrier between them.
- 2 Adaptive radiation has taken place due to the many available ecological niches found in the crater lakes.
- 3 The 11 species of cichlids within Barombi Mbo arose by sympatric speciation due to reproductive isolation.

A 1 only **B** 1 and 2 only **C** 2 and 3 only **D** 1, 2 and 3

- 28** Two individuals took part in a study to investigate the effectiveness of two different types of immunisation. Individual S received an injection of antibodies against tetanus and Individual T received a tetanus vaccination.

Which of the options below shows correctly the changes to the antibody concentration in the blood of S and T?



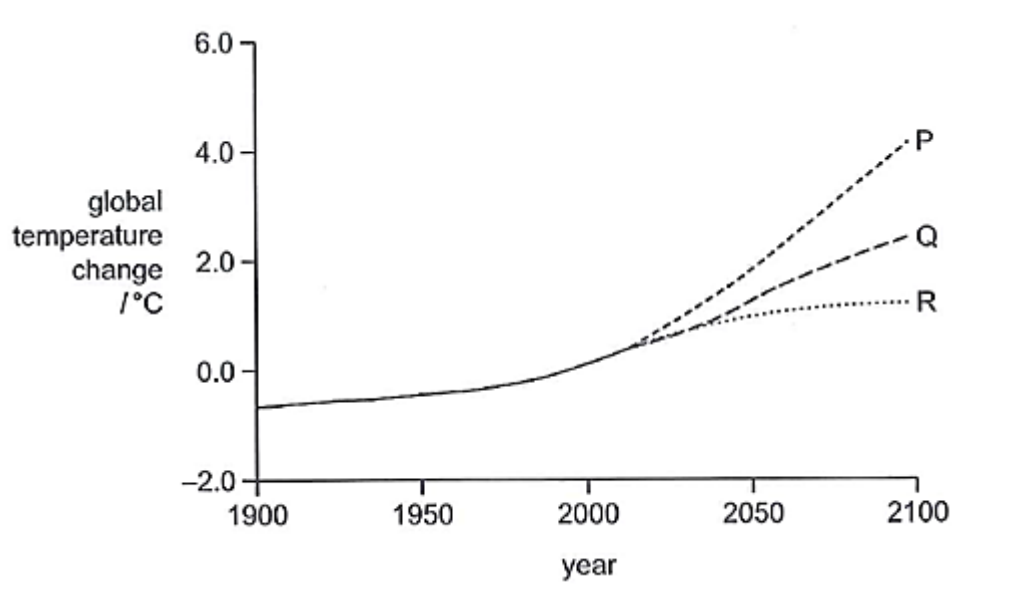
- 29** Global warming caused by the enhanced greenhouse effect is likely to have major consequences for arctic ecosystems. Which of the following are likely to occur in the arctic if the Earth's surface temperature rises?

- I. Decreased rates of decomposition of detritus
- II. Increased range of predators from temperate regions
- III. Increase in numbers of pest species and pathogens

- A** I and II only
B I and III only
C II and III only
D I, II and III

- 30 The graph shows the predicted change in global temperatures using three different models, **P**, **Q** and **R**.

Model **Q** assumes that no new factors act to influence the rate of climate change.



The predictions of models **P** and **R** can be explained using some of the following statements.

- 1 Expansion of rainforest.
- 2 Increase in frequency of forest fires.
- 3 Permanently frozen soil and sediment in the Arctic begin to thaw.
- 4 Rising sea temperatures will reduce the solubility of greenhouse gases in the oceans.

Which of these statements support prediction of models **P** and **R**?

	Statements that support prediction P	Statements that support prediction R
A	2, 3 and 4	1
B	1 and 3	2 and 4
C	2	1, 3 and 4
D	3 and 4	1 and 2